

Brachyolmia–Amelogenesis Imperfecta Syndrome (Dental Anomalies and Short Stature Syndrome, DASS): A Comprehensive Disease Report

Disease: Brachyolmia–Amelogenesis Imperfecta Syndrome **Primary synonym:** Dental Anomalies and Short Stature syndrome (DASS) **Causal gene:** *LTBP3* (Latent TGF- β Binding Protein 3) **Category:** Mendelian, autosomal recessive **Key identifiers:** OMIM 601216 (disease phenotype); OMIM 602090 (*LTBP3* gene); HGNC:6716; NCBI Gene 4054; UniProt Q9NS15

Summary

Brachyolmia–Amelogenesis Imperfecta Syndrome — now most often called **Dental Anomalies and Short Stature syndrome (DASS; OMIM 601216)** — is an **ultra-rare autosomal recessive Mendelian disorder** caused by **biallelic loss-of-function (hypomorphic) variants in *LTBP3***, the gene encoding Latent Transforming Growth Factor- β Binding Protein 3. First delineated by Verloes and colleagues in 1996 as a new form of skeletal dysplasia combining amelogenesis imperfecta with platyspondyly, the molecular cause was established in 2015 when whole-exome sequencing of four families identified recessive deletion, nonsense, and splice-site *LTBP3* mutations. The disorder is defined by a characteristic clinical triad: **significant short stature with brachyolmia** (a mild, generalized platyspondylic skeletal dysplasia), **hypoplastic amelogenesis imperfecta with near-absent enamel**, and a spectrum of **orodental anomalies** (oligodontia/hypodontia, delayed or failed eruption, taurodontism, abnormal dentin, underdeveloped maxilla).

Mechanistically, *LTBP3* is a secreted extracellular-matrix protein that anchors the large latent TGF- β complex (TGF- β + LAP propeptide + LTBP) to fibrillin-1 microfibrils, thereby governing the secretion, matrix localization, and bioavailability of TGF- β . Biallelic *LTBP3* loss disrupts assembly of the TGF- β –LAP–*LTBP3* latent complex, perturbing TGF- β secretion/activation and downstream SMAD2/3 and ERK1/2 signaling in a **context- and dose-dependent** manner;

producing dental, skeletal, and cardiovascular consequences. The *Ltbp3*-null mouse faithfully recapitulates the human phenotype (reduced body size, craniofacial/skull-base synchondrosis abnormalities, high bone mass with low turnover, and thin-to-absent enamel), providing strong causal validation.

Clinically, DASS is a chronic, lifelong, largely **non-life-threatening** condition dominated by dental and skeletal morbidity, but it carries important **cardiovascular risk** — **thoracic aortic aneurysm and dissection (TAAD)** has been documented in both biallelic and heterozygous *LTBP3* carriers. *LTBP3* also exhibits a striking **allelic series**: monoallelic missense or de novo variants instead cause dominant **acromicric dysplasia** or lethal **geleophysic dysplasia** (respiratory failure in early childhood), placing the gene within the acromelic/microfibrillar-network disorder family. There is **no disease-modifying therapy**; management is supportive and multidisciplinary (restorative dentistry with ceramic crowns as first-line, orthodontic/orthognathic/prosthetic rehabilitation, growth monitoring, and echocardiographic aortic surveillance), while prevention is reproductive (genetic counseling, carrier/cascade testing, prenatal and preimplantation genetic diagnosis), particularly relevant in the consanguineous and founder populations in which the disorder is enriched.

1. Disease Information

Overview. DASS is a rare autosomal recessive syndrome combining a mild spondylar skeletal dysplasia (brachyolmia) with a severe enamel defect (hypoplastic amelogenesis imperfecta) and short stature. The condition was first characterized clinically by Verloes et al. (1996) in two children of consanguineous parents and molecularly resolved to *LTBP3* by Huckert et al. (2015). It is described as being "characterized by significant short stature with brachyolmia and hypoplastic amelogenesis imperfecta (AI) with almost absent enamel" ([PMID: 25669657](#)).

Key identifiers.

Resource	Identifier
OMIM (disease)	601216
OMIM (gene, <i>LTBP3</i>)	602090
HGNC	HGNC:6716
NCBI Gene	4054
UniProt	Q9NS15
MONDO	Brachyolmia–amelogenesis imperfecta syndrome / DASS (cross-referenced to OMIM 601216)

Synonyms and alternative names: - Dental Anomalies and Short Stature syndrome (DASS) — the current preferred designation - Brachyolmia–amelogenesis imperfecta syndrome - Brachyolmia with amelogenesis imperfecta - Amelogenesis imperfecta and platyspondyly (as originally described) - LTBP3-related skeletal dysplasia

The synonym "DASS" and OMIM number are established explicitly: "Mutations in LTBP3 are associated with Dental Anomalies and Short Stature syndrome (DASS; MIM 601216)" ([PMID: 35352826](#)).

Source of information. The evidence base is derived from **aggregated disease-level resources** — case reports and small family series (whole-exome sequencing studies of consanguineous families), a mouse knockout model, and biochemical studies of TGF- β latent complex biology — rather than from large individual-patient EHR datasets. As of 2020 only ~20 individuals from 9 families had been reported.

2. Etiology

Disease causal factors. DASS is a **monogenic (Mendelian) genetic disorder**. The primary and sole established cause is **biallelic loss-of-function / hypomorphic variation in *LTBP3***. There is no environmental, infectious, or acquired etiology. Huckert et al. identified "recessive hypomorphic mutations including deletion, nonsense and splice mutations, in the LTBP3 gene, which is involved in the TGF-beta signaling pathway" ([PMID: 25669657](#)).

Genetic risk factors. - **Causal variants:** Biallelic (homozygous or compound heterozygous) loss-of-function *LTBP3* variants — including deletions, nonsense, frameshift, and canonical splice-site mutations. - **Consanguinity:** A major risk factor. Most reported families are consanguineous, increasing the probability of homozygosity for a rare recessive allele ([PMID: 8721563](#)). - **Founder alleles:** Population-specific founder variants exist — e.g., a Druze Arab founder variant (c.1346-1G>A) localized by homozygosity mapping to chromosome 11p11.2–q13.3 ([PMID: 37394436](#)).

Environmental risk factors. None identified. As a fully penetrant recessive Mendelian disorder, no environmental, occupational, or lifestyle exposures are known to cause or trigger DASS.

Protective factors. No genetic modifier alleles or environmental protective factors are documented (see Modifier Genes below).

Gene–environment interactions. None documented. Given the monogenic recessive architecture, disease occurrence is determined by genotype; however, mechanical/environmental factors may plausibly modulate the *severity* of downstream consequences (e.g., spinal deformity influencing aortic mechanics — see Prognosis), though this is inferred rather than demonstrated in patients.

3. Phenotypes

The phenotype is remarkably consistent across families and populations. Below, phenotypes are grouped with suggested HPO terms, characteristics, and frequency.

Phenotype	Type	HPO term (suggested)	Onset	Severity	Frequency
Hypoplastic amelogenesis imperfecta / near-absent enamel	Physical/ dental sign	HP:0000705 (Amelogenesis imperfecta)	Congenital (both dentitions)	Severe	Near-universal (defining)
Short stature	Physical sign	HP:0004322 (Short stature)	Postnatal/ childhood	Moderate–severe	Near-universal (defining)
Brachyolmia / platyspondyly	Skeletal/ radiographic	HP:0000926 (Platyspondyly)	Childhood	Mild–moderate	Near-universal (defining)
Oligodontia / hypodontia	Dental	HP:0000670 / HP:0000668	Congenital	Variable	Frequent
Delayed/failed tooth eruption	Dental	HP:0000684 (Delayed eruption of teeth)	Childhood	Variable	Frequent
Taurodontism	Dental	HP:0000679 (Taurodontism)	Congenital	Mild	Reported
Abnormal dentin	Dental	HP:0011063 (Abnormal dentin morphology)	Congenital	Variable	Reported
Underdeveloped/hypoplastic maxilla	Craniofacial	HP:0000327 (Hypoplasia of the maxilla)	Childhood	Variable	Frequent
Broad femoral necks	Skeletal	HP:0012865 (Broad femoral neck)	Childhood	Mild	Reported
Vertebral scalloping / rectangular vertebrae	Radiographic	Posterior vertebral scalloping	Childhood	Mild	Reported

Phenotype	Type	HPO term (suggested)	Onset	Severity	Frequency
Thoracic aortic aneurysm/ dissection	Cardiovascular	HP:0004942 / HP:0002647	Adult (variable)	Severe (serious complication)	Reported subset
Nephrocalcinosis	Laboratory/ imaging	HP:0000121 (Nephrocalcinosis)	Variable	Variable	Occasional (Moroccan families)
Malocclusion / open bite	Dental	HP:0000689 (Dental malocclusion)	Childhood	Variable	Frequent

The original description captured the skeletal-dental core: "amelogenesis imperfecta (absence of the enamel cap) associated with brachyolmia-like anomalies: platyspondyly with short pedicles, narrow intervertebral and interpedicular distances, rectangular-shaped vertebrae with posterior scalloping and herniation of the nuclei, and broad femoral necks" ([PMID: 8721563](#)). Later series expanded the orodental spectrum and added aortic involvement: "hypoplastic type amelogenesis imperfecta, hypodontia, underdeveloped maxilla, short stature, brachyolmia, aneurysm and dissection of the thoracic aorta" ([PMID: 35352826](#)).

Quality-of-life impact. The amelogenesis imperfecta component drives substantial functional and psychosocial burden. In a cohort of 68 children/adolescents (7–19 y), "Children under 13 years were more frequently reported functional difficulties, such as pain and eating challenges, while adolescents (≥ 13 years) more often expressed psychosocial concerns including bullying and low self-confidence" ([PMID: 42113459](#)). Interview studies find "the impact of AI on quality of life is more severe than previously appreciated" ([PMID: 38909645](#)). Short stature and skeletal features add further functional and psychosocial dimensions.

4. Genetic / Molecular Information

Causal gene. *LTBP3* (Latent TGF- β Binding Protein 3), chromosome 11q13.1; gene OMIM 602090; HGNC:6716; NCBI Gene 4054; protein UniProt Q9NS15.

Pathogenic variants. DASS is caused by **biallelic loss-of-function/hypomorphic** variants. Reported variant classes include:

Variant (example)	Type	Predicted effect	Population/Source
Deletion, nonsense, splice mutations	LOF	Loss of function	Four families (PMID: 25669657)
c.2495delT (p.Phe832SerfsTer36)	Frameshift	LOF	Moroccan family 1 (PMID: 35998423)
c.3716G>A (p.Cys1239Tyr)	Missense (destabilizing)	Structural destabilization	Moroccan family 2 (PMID: 35998423)
c.625dup (p.Leu209fs) + c.1965del (p.Arg656fs)	Compound het frameshift	LOF	TAAD patient with short stature/dental problems (PMID: 34906192)
c.1346-1G>A	Splice acceptor	LOF	Druze Arab founder (PMID: 37394436)

ACMG/AMP classification. Reported causal variants are generally **pathogenic/likely pathogenic** (null variants meeting PVS1-type criteria, segregating in consanguineous families with a specific, well-established phenotype). The missense p.Cys1239Tyr variant was supported functionally by molecular modeling that "disclosed a possible destabilization of the wild-type structure" (PMID: 35998423).

Allele frequency. Causal alleles are ultra-rare/private; population-database (gnomAD) frequencies for reported pathogenic variants are extremely low, consistent with a recessive ultra-rare disorder enriched by consanguinity and founder effects.

Somatic vs germline. All disease-causing variants are **germline**; no somatic contribution is relevant.

Functional consequences. Loss of function — variants abrogate LTBP3's ability to form the TGF- β -LAP-LTBP3 latent complex and localize latent TGF- β to fibrillin microfibrils: "failure of TGF β -LAP-LTBP3 complex formation, and subsequent disruption of TGF β secretion and activation" (PMID: 35352826).

Genotype-phenotype correlation (allelic series). A key principle: "biallelic loss-of-function mutations cause DASS, monoallelic missense" variants cause acromicric dysplasia (PMID: 30887145). Dominant missense (c.2087C>G, p.Ser696Cys) causes acromicric dysplasia, and de novo heterozygous variants (splice c.1846+5G>A; stop-loss p.1304ext12) cause lethal geleophysic dysplasia (PMID: 27068007).

Modifier genes. No specific modifier genes are identified for DASS. Given LTBP3's function within the fibrillin-microfibril/TGF- β network, genes encoding partner proteins (FBN1, LTBP2, ADAMTS10/17, ADAMTSL2) are biologically plausible modifiers but not demonstrated.

Epigenetic information. No DNA-methylation, histone-modification, or chromatin-based mechanisms have been reported for DASS. Not applicable based on current evidence.

Chromosomal abnormalities. DASS is a single-gene disorder; no aneuploidy, translocation, or large structural rearrangement is characteristic. Homozygosity mapping (not a structural abnormality) identified the 11p11.2–q13.3 founder region in Druze Arab patients ([PMID: 37394436](#)).

5. Environmental Information

Environmental factors, lifestyle factors, and infectious agents: Not applicable. DASS is a purely genetic, autosomal recessive Mendelian disorder. No toxins, radiation, pollution, occupational exposures, dietary/behavioral factors, or infectious agents contribute to its causation. Consanguinity (a demographic/social factor) increases the probability of an affected offspring but is not an environmental cause of the molecular lesion itself.

6. Mechanism / Pathophysiology

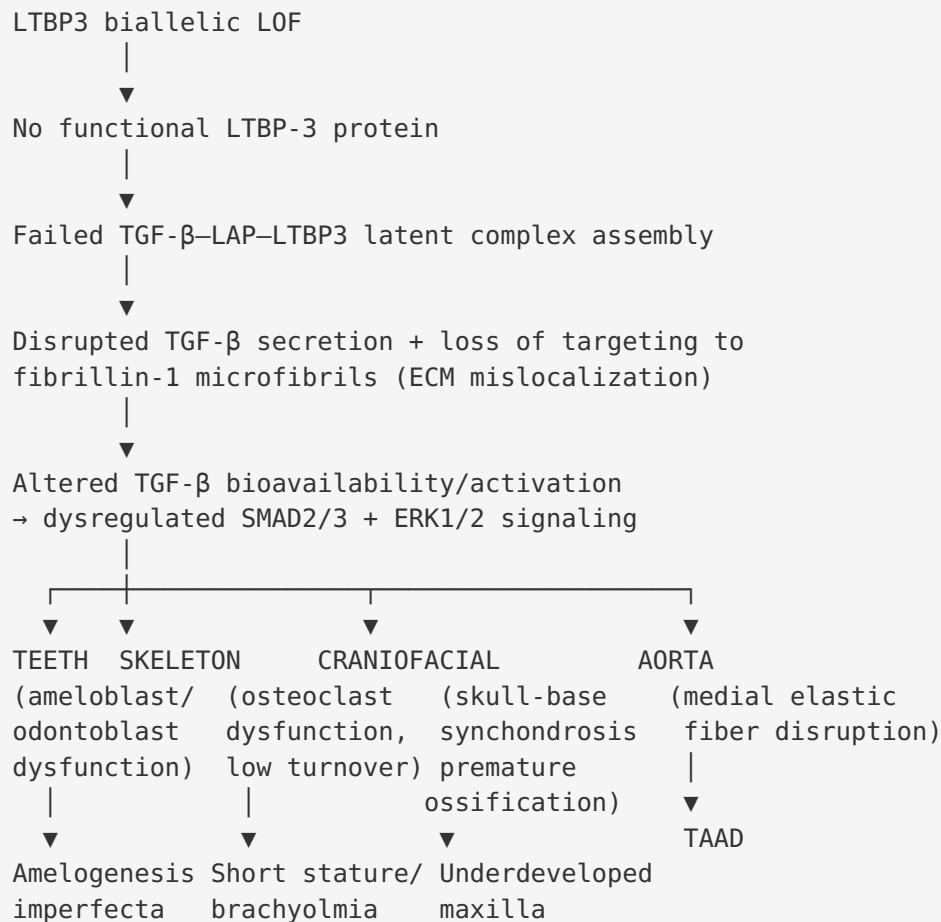
Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function *LTBP3* variants** (deletion/nonsense/frameshift/splice) **lead to** absent or non-functional LTBP-3 protein. (*Demonstrated.*)
2. Loss of functional LTBP-3 **results in** failure to assemble the large latent complex (TGF- β + LAP + LTBP-3), i.e., "failure of TGF β -LAP-LTBP3 complex formation" ([PMID: 35352826](#)). (*Demonstrated biochemically.*)
3. Failed complex assembly **disrupts** TGF- β secretion and its targeting to fibrillin-1 microfibrils in the ECM (matrices lacking fibrillin-1 also lack LTBP-3). (*Demonstrated in vitro.*)
4. Mislocalized/reduced latent TGF- β **alters** the local bioavailability and activation of TGF- β , **dysregulating** downstream SMAD2/3 and ERK1/2 signaling in a **context- and dose-**

dependent manner ([PMID: 26494287](#)). (*Demonstrated in mouse aorta; inferred for other tissues.*)

5. **Branch A — Teeth:** In differentiated ameloblasts (which express *Ltbp3*) and odontoblasts, disrupted TGF- β signaling **impairs** enamel matrix deposition and mineralization, **producing** hypoplastic amelogenesis imperfecta with near-absent enamel and abnormal dentin. (*Demonstrated: mouse enamel phenotype + ameloblast/odontoblast expression.*)
6. **Branch B — Skeleton:** Reduced TGF- β in bone and cartilage **compromises** osteoclast function and **decreases** bone turnover, **causing** premature ossification of skull-base synchondroses, altered vertebral/long-bone growth, short stature, brachyolmia, and (in mouse) an osteopetrosis-like high-bone-mass state ([PMID: 15878314](#)). (*Demonstrated in mouse; inferred in humans.*)
7. **Branch C — Cardiovascular:** Altered TGF- β regulation and ECM/microfibril integrity **predispose** the thoracic aortic wall to medial elastic-fiber disruption, **leading to** aneurysm and dissection (TAAD) ([PMID: 34906192](#)). (*Demonstrated in patients and mouse models; context-dependent.*)
8. These branches **manifest clinically** as the DASS triad plus its cardiovascular complications.

Mechanistic detail



- **Molecular pathways:** TGF- β signaling (canonical SMAD2/3 and non-canonical ERK1/2/MAPK). "TGF β is secreted from cells as a latent complex consisting of TGF β , the TGF β propeptide, and a molecule of latent TGF β binding protein (LTBP)" (PMID: 26494287).
- **Cellular processes:** Ameloblast enamel-matrix secretion, odontoblast dentinogenesis, osteoclast-mediated bone resorption/turnover, chondrocyte/synchondrosis ossification, aortic smooth-muscle/ECM homeostasis.
- **Protein dysfunction:** Loss of function of a secreted ECM scaffolding protein; missense variants (e.g., p.Cys1239Tyr; EGF-like calcium-binding domain variants) can act via structural destabilization or dominant-negative effects in the allelic-series disorders.
- **Tissue damage mechanisms:** Aortic medial "disruption and fragmentation of medial elastic fibers" (PMID: 26494287); defective enamel biomineralization; abnormal endochondral/synchondrosis ossification.

- **Biochemical abnormality:** Failure of latent TGF- β complex assembly and reduced ECM-localized TGF- β .

Suggested GO terms: transforming growth factor beta receptor signaling pathway (GO:0007179); regulation of transforming growth factor beta production (GO:0071634); extracellular matrix organization (GO:0030198); biomineral tissue development (GO:0031214); bone resorption (GO:0045453); ossification (GO:0001503). **Suggested CL terms:** ameloblast (CL:0000059); odontoblast (CL:0000060); osteoclast (CL:0000092); osteoblast (CL:0000062); chondrocyte (CL:0000138); vascular smooth muscle cell (CL:0000359).

7. Anatomical Structures Affected

Organ level (primary): Teeth (enamel and dentin), axial skeleton (vertebrae/spine), long bones, craniofacial skeleton (skull-base synchondroses, maxilla). **Secondary/complication organs:** Thoracic aorta and cardiovascular structures (interatrial septum, cardiac valves); occasionally kidneys (nephrocalcinosis in some Moroccan families, [PMID: 35998423](#)). **Body systems:** Skeletal, dental/oral, cardiovascular; (renal, occasionally).

Tissue and cell level: Mineralized dental tissues (enamel produced by ameloblasts; dentin by odontoblasts) — "Differentiated ameloblasts synthesizing enamel matrix proteins and odontoblasts expressed the gene" ([PMID: 25669657](#)); cartilage/bone (chondrocytes, osteoblasts, osteoclasts); aortic media (elastic fibers, vascular smooth muscle cells).

Subcellular level: LTBP-3 is a **secreted extracellular matrix protein** localized to **fibrillin-1 microfibrils** (extracellular region; GO:0031012 extracellular matrix). It transits the secretory pathway (ER/Golgi) prior to secretion.

Localization (UBERON): tooth enamel (UBERON:0001752); dentine (UBERON:0001751); vertebral column (UBERON:0001130); femur/femoral neck (UBERON:0000981); maxilla (UBERON:0002397); cranial base region (UBERON:0011156); thoracic aorta (UBERON:0001515).

Lateralization: Bilateral/generalized (systemic skeletal and dental involvement); aortic disease affects the midline thoracic aorta.

8. Temporal Development

Onset: Congenital to early childhood. Enamel defects affect both primary and permanent dentitions (congenital), and short stature/brachyolmia become apparent in the postnatal/childhood growth period. Onset pattern is **chronic and insidious** (a developmental dysplasia), not acute.

Progression: DASS is a **chronic, lifelong, essentially non-progressive skeletal dysplasia**. The skeletal and dental features are developmentally determined and stable rather than degenerative. Dental morbidity (enamel breakdown, caries susceptibility, tooth loss) can accumulate over time if untreated. The **cardiovascular complication (TAAD) is a later, potentially progressive risk** that can present in adulthood and represents the most serious temporal dimension.

Disease course pattern: Stable/chronic for skeletal-dental features; the aortic component is progressive/episodic (aneurysm growth punctuated by acute dissection risk). Disease duration is lifelong.

Patterns / critical periods: - **Odontogenesis (fetal–childhood):** the critical window during which enamel/dentin defects are established — no post-hoc biological remediation of enamel is possible. - **Growth period (childhood–adolescence):** window for growth monitoring and orthodontic/orthognathic planning. - **Adulthood:** window for aortic surveillance and timely intervention. No spontaneous remission occurs; "remission" applies only to symptom control via restorative treatment.

9. Inheritance and Population

Epidemiology. DASS is **ultra-rare**: "Only 20 individuals from nine families have been previously reported, with a consistent phenotype of short stature, brachyolmia, and amelogenesis imperfecta" ([PMID: 32432408](#)). Additional families have since been reported across French, Turkish, Moroccan, Indian, Druze Arab, and East Asian populations. Precise prevalence/incidence figures are not established given the rarity; the disorder falls well below the 1/1,000,000 range typical of ultra-rare recessive dysplasias.

Inheritance pattern: Autosomal recessive — "Inheritance appears to be autosomal recessive" ([PMID: 8721563](#)); caused by biallelic hypomorphic/LOF variants.

Penetrance and expressivity: The core triad appears **highly/completely penetrant** in individuals with biallelic LOF variants. **Expressivity is variable**, particularly for severity of dental involvement, nephrocalcinosis (some families), and cardiovascular risk. Reports note "difference in severity" even within a family ([PMID: 35998423](#)).

Genetic anticipation: Not applicable (not a repeat-expansion disorder).

Germline mosaicism: Not specifically reported.

Founder effects: Documented — the Druze Arab founder variant c.1346-1G>A, with homozygosity mapping to chromosome 11p11.2–q13.3 ([PMID: 37394436](#)).

Consanguinity: A major contributor; most families are consanguineous ([PMID: 8721563](#)).

Carrier frequency: Not established at the population level; expected to be very low outside founder groups.

Population demographics / geographic distribution: Reported worldwide but enriched in populations with high consanguinity rates and in specific founder communities. **Sex ratio** is expected to be **1:1** (autosomal recessive; no sex bias reported). **Age distribution:** affected individuals identified from childhood (dental/growth features) through adulthood (cardiovascular presentation).

10. Diagnostics

Clinical recognition. Diagnosis begins with recognition of the characteristic triad — **short stature + brachyolmia (platyspondyly) + hypoplastic amelogenesis imperfecta** — on combined clinical, dental, and radiographic examination.

Imaging: Skeletal radiographs of the **spine and pelvis** reveal platyspondyly, short pedicles, narrow intervertebral/interpedicular distances, rectangular vertebrae with posterior scalloping, and broad femoral necks ([PMID: 8721563](#)). Dental radiographs demonstrate near-absent enamel, taurodontism, oligodontia/hypodontia, and unerupted teeth. **Echocardiography** (and cross-sectional aortic imaging) is indicated to evaluate for thoracic aortic aneurysm and cardiac structural anomalies.

Genetic testing (definitive). Molecular confirmation is by identifying **biallelic *LTBP3* variants**. Recommended approaches: - **Whole-exome sequencing (WES):** the primary discovery and diagnostic tool in reported families ([PMID: 25669657](#), [PMID: 35998423](#)). - **Targeted single-gene *LTBP3* sequencing** or **gene panels** (skeletal dysplasia / amelogenesis

imperfecta panels). - **Whole-genome sequencing (WGS)** where WES is uninformative. - **Homozygosity mapping / chromosomal microarray (SNP array):** especially valuable in consanguineous families to identify runs of homozygosity harboring *LTBP3* — "One homozygote region in chromosome 11 (11p11.2-11q13.3) was found in all patients" ([PMID: 37394436](#)). - Karyotyping, FISH, mitochondrial DNA testing, and repeat-expansion testing are **not applicable**.

Omics-based diagnostics: Not routinely used; research-level functional studies (e.g., molecular modeling of missense variants, TGF- β signaling assays) support variant classification.

Clinical criteria / differential diagnosis. No formal consensus criteria exist; diagnosis is triad-based plus molecular confirmation. Key **differential diagnoses:**

Condition	Gene	Distinguishing feature
Brachyolmia, dominant type	<i>TRPV4</i>	Dominant; no amelogenesis imperfecta
Brachyolmia, recessive (Hobaek/Maroteaux) / PAPSS2 type	<i>PAPSS2</i>	Recessive brachyolmia without the AI/enamel defect
Acromicric dysplasia	<i>LTBP3</i> (monoallelic missense), <i>FBN1</i> , <i>ADAMTSL2</i>	Dominant; short stature with stubby hands, no AI
Geleophysic dysplasia	<i>LTBP3</i> (de novo), <i>FBN1</i> , <i>ADAMTSL2</i> , <i>ADAMTS10/17</i>	Lethal cardiorespiratory disease; "happy face"; no AI triad
Isolated amelogenesis imperfecta	<i>AMELX</i> , <i>ENAM</i> , <i>MMP20</i> , <i>FAM83H</i> , etc.	Enamel defect without brachyolmia/short stature

Brachyolmia's genetic heterogeneity is well established: "there are 3 and possibly 4 different types of brachyolmia" ([PMID: 2669482](#)).

Screening. Cascade carrier testing within families and founder-population carrier screening are the principal screening modalities (see Prevention).

11. Outcome / Prognosis

Overall prognosis: DASS is a **chronic, lifelong, generally non-life-threatening** condition. The dominant burden is **dental** (pain, hypersensitivity, tooth breakdown/loss, malocclusion, aesthetic/psychosocial impact) and **short stature/skeletal** — none of which are inherently life-limiting.

Serious/lethal risks: - **Thoracic aortic aneurysm and dissection (TAAD):** the principal life-threatening complication, reported in biallelic and heterozygous *LTBP3* carriers — "The identification of *LTBP3* mutations in TAAD patients in our study provided more clinical evidence to support its association with TAAD" ([PMID: 34906192](#)). Associated cardiac findings include interatrial septal aneurysm, ASD, and tricuspid valve prolapse. - **Respiratory failure (severe allelic-series end):** at the geleophysic dysplasia end of the *LTBP3* spectrum, de novo heterozygous variants caused "two unrelated GD individuals who had died in early childhood from respiratory failure" ([PMID: 27068007](#)) — a distinct, more severe *LTBP3*-related disorder, not classic DASS, but defining the lethal extreme of the gene's phenotypic range.

Mechanistic prognostic insight: Spinal deformity may adversely influence aortic biomechanics — "a spinal deformity either remains or is exacerbated in the absence of *LTBP-3* and seems to adversely affect the axial mechanical properties of the thoracic aorta" ([PMID: 30306291](#)).

Morbidity, disability, and quality of life: Chronic dental morbidity and psychosocial impact predominate; functional difficulties (pain, eating) and psychosocial concerns (bullying, low self-confidence) are age-dependent ([PMID: 42113459](#)). With restorative dental care, satisfaction and function improve markedly.

Prognostic factors: presence and rate of aortic dilatation; severity of dental involvement; skeletal deformity. **Prognostic biomarkers:** none validated; aortic diameter on serial imaging is the practical prognostic marker for cardiovascular risk.

12. Treatment

No disease-modifying/curative or gene-directed therapy exists. Management is symptomatic and multidisciplinary.

Dental / restorative (first-line for the AI component): - **Single-tooth ceramic crowns** are now recommended as first choice with high success across AI types — "single-tooth ceramic crowns should be the first choice of treatment" (PMID: 38909645). - **Resin composite restorations** for mild/hypoplastic cases. - Restorative therapy relieves symptoms — "In young patients with AI symptoms of pain and hypersensitivity decreased, and aesthetics were improved following all types of restorative therapy" (PMID: 38909645). - Severe cases: "AI is frequently accompanied by unesthetic appearance, open bite deformity and malocclusion, a multidisciplinary approach is often required" — combined orthodontic + orthognathic surgical + prosthodontic rehabilitation over several years (PMID: 23811667). - Suggested NCIT: dental restoration procedure; dental crown; orthognathic surgery; prosthodontic rehabilitation.

Cardiovascular: Echocardiographic/imaging **surveillance** for thoracic aortic aneurysm; standard aneurysm management (blood-pressure control, activity guidance, and surgical repair when indicated) is warranted given the documented TAAD risk (PMID: 34906192).

Growth/endocrine: Growth monitoring; management of short stature is supportive.

Supportive/rehabilitative: Pain management, nutrition support (eating difficulties), and psychosocial support for the QoL burden.

Pharmacogenomics, gene therapy, cell therapy, RNA-based therapy, targeted therapy, immunotherapy: None available/applicable at present.

Experimental treatments: No disease-specific registered clinical trials identified. Given the TGF- β mechanism, TGF- β pathway modulation is a conceptual (unproven) avenue.

Personalized medicine: Care is tailored to individual phenotype severity (dental, skeletal, cardiovascular), but no genotype-guided pharmacotherapy exists.

13. Prevention

No **primary prevention** of the underlying genetic cause is possible. Prevention is **reproductive and secondary/tertiary**.

Reproductive/genetic prevention: - **Genetic counseling** for affected families, emphasizing the 25% recurrence risk in autosomal recessive inheritance. - **Carrier and cascade testing**, especially in consanguineous families and founder populations — carrier-state evaluation "in the particular community" is a documented strategy (PMID: 37394436). - **Prenatal diagnosis** and **preimplantation genetic testing (PGT)** for known familial *LTBP3* variants. The pathway

is illustrated in analogous AR consanguineous skeletal dysplasias: "Both parents were heterozygous carriers. Following genetic counseling, the family opted for pregnancy termination" (PMID: 40368527).

Secondary/tertiary prevention: - Early dental restorative intervention to prevent tooth breakdown, pain, and secondary caries. - Echocardiographic aortic surveillance to enable timely intervention and **prevent aortic dissection**. - Orthodontic/orthognathic planning to prevent progressive malocclusion complications.

Immunization, public-health, and environmental interventions: Not applicable (non-infectious, non-environmental genetic disorder).

14. Other Species / Natural Disease

Taxonomy / model species: *Mus musculus* (NCBI Taxon 10090) is the principal model species. **Orthologous gene:** mouse *Ltbp3* (NCBI Gene ID 16997) is orthologous to human *LTBP3* (NCBI Gene ID 4054); the protein is evolutionarily conserved across vertebrates. **Natural disease in other species: None catalogued.** No naturally occurring *LTBP3*-equivalent disease has been reported in companion animals or livestock (e.g., in OMIA). Veterinary relevance is therefore currently nil. **Comparative biology:** The *Ltbp3*-null mouse demonstrates strong cross-species conservation of the disease mechanism (skeletal, craniofacial, and enamel phenotypes recapitulated). **Transmission / zoonotic potential:** Not applicable (non-infectious genetic disorder). **Breed (VBO):** Not applicable.

15. Model Organisms

Principal model: the constitutive ***Ltbp3*-knockout (null) mouse** ("we generated *Ltbp3* null mice," PMID: 15878314). Model type: mammalian, genetic knockout (MGI).

Phenotype recapitulation:

Human DASS feature	Mouse <i>Ltbp3</i> -null phenotype	Source
Short stature	Reduced body size	PMID: 15878314
Craniofacial/skull-base anomalies	Early ossification of skull-base synchondroses; craniofacial abnormalities	PMID: 15878314
Altered bone	Osteopetrosis-like high bone mass; decreased bone turnover; persistent cartilage remnants	PMID: 15878314
Amelogenesis imperfecta	Very thin to absent enamel in incisors and molars	PMID: 25669657

"the mutant mice displayed very thin to absent enamel in both incisors and molars, hereby recapitulating the AI phenotype in the human disorder" ([PMID: 25669657](#)). The skeletal mechanism was proposed as: "lack of *Ltbp3* results in decreased levels of TGF- β in bone and cartilage, which leads to compromised osteoclast function and decreased bone turnover" ([PMID: 15878314](#)).

Mechanistic model use: The *Ltbp3*-null mouse, crossed into fibrillin-1-deficient (Marfan) mice, has been central to dissecting LTBP3's role in TGF- β -driven aortic disease — "we genetically suppressed *Ltbp3* expression in a mouse model of progressively severe MFS" ([PMID: 26494287](#)). Strikingly, in that context "MFS mice lacking LTBP-3 have improved survival, essentially no aneurysms, reduced disruption and fragmentation of medial elastic fibers, and decreased Smad2/3 and Erk1/2 activation in their aortas" ([PMID: 26494287](#)).

Model limitations: The knockout is a complete null, whereas human DASS arises from hypomorphic/LOF alleles that may retain residual function; species differences in tooth continuous growth (mouse incisors) and skeletal proportion limit direct translation of some features; the human cardiovascular (TAAD) phenotype is best studied in sensitized (Marfan) backgrounds rather than in *Ltbp3*-null mice alone.

Applications: Study of enamel biomineralization, skull-base/vertebral development, bone turnover, and TGF- β regulation in aortic biology. **Other model systems:** In vitro biochemical studies of latent-complex assembly and fibrillin-1 microfibril targeting complement the mouse ([PMID: 26494287](#)).

Mechanistic Model / Interpretation

DASS is best understood as a **TGF- β "delivery/localization" disorder**. LTBP-3 does not itself signal; it is a molecular chaperone/scaffold that secures the latent TGF- β complex to the fibrillin-1 microfibrillar network of the extracellular matrix. When LTBP-3 is absent (biallelic LOF), latent TGF- β is not properly secreted or deposited in the ECM, so its **spatiotemporal availability for activation is disturbed**. Because TGF- β is a master regulator of skeletal, dental, and vascular ECM homeostasis, a single upstream lesion fans out into three phenotypic branches — enamel/dentin (ameloblast/odontoblast dysfunction), the growth skeleton (osteoclast-driven low bone turnover, premature synchondrosis ossification), and the aortic wall (elastic-fiber fragmentation $\rightarrow$ aneurysm).

Crucially, the *LTBP3* **allelic series** demonstrates that gene dosage and variant mechanism dictate the phenotype:

Genotype / mechanism	Disorder	Inheritance	Severity
Biallelic loss of function	DASS (brachyolmia–AI)	Autosomal recessive	Chronic; dental/skeletal; aortic risk
Monoallelic missense (e.g., p.Ser696Cys)	Acromicric dysplasia	Autosomal dominant	Short stature, no AI
De novo heterozygous (splice / stop-loss)	Geleophysic dysplasia	Sporadic/ dominant	Lethal (early respiratory failure)

This dose/mechanism dependence mirrors the paradoxical mouse aortic data, in which *removing* *Ltbp3* in a Marfan background *improves* aortic disease — underscoring that LTBP-3's net effect on TGF- β signaling is **context-dependent**, promoting disease in some tissues while its loss drives disease in others.

Evidence Base

PMID	Title (abbrev.)	Role in this report
25669657	<i>LTBP3 mutations cause brachyolmia with AI</i>	Landmark gene-discovery paper; defines triad, causal LOF variants, mouse enamel recapitulation, ameloblast/odontoblast expression
35352826	<i>Expanding genotypic/phenotypic spectrums of LTBP3 in DASS</i>	DASS name + OMIM 601216; latent-complex failure mechanism; aortic features
30887145	<i>Genotype-phenotype correlation in LTBP3 disorders</i>	Core biallelic-LOF-vs-monoallelic-missense principle
27068007	<i>LTBP3 in acromicric and geleophysic dysplasia</i>	Allelic series; lethal respiratory outcome; microfibrillar-network positioning
15878314	<i>Osteopetrosis-like phenotype in Ltbp3-deficient mice</i>	Mouse model; skeletal/craniofacial recapitulation; TGF- β /osteoclast mechanism
26494287	<i>LTBP-3 contribution to thoracic aneurysm in Marfan</i>	Latent-complex biology; SMAD2/3 + ERK1/2; context-dependent aortic role
34906192	<i>Novel LTBP3 mutations in TAAD</i>	Links biallelic LTBP3 loss to thoracic aortic aneurysm/dissection
8721563	<i>New skeletal dysplasia with AI and platyspondyly</i>	Original clinical delineation; AR inheritance; skeletal hallmarks
32432408	<i>Bi-allelic LTBP3 variants — first Indian patient</i>	Ultra-rarity (20 individuals/9 families); consistent phenotype
35998423	<i>LTBP3 variants in two Moroccan families</i>	Novel variants; nephrocalcinosis; intra-familial severity variation
37394436	<i>LTBP3 variant in Druze Arab patients</i>	Founder variant; homozygosity mapping; carrier-screening strategy
2669482	<i>Brachyolmia heterogeneity</i>	Differential diagnosis; genetic heterogeneity of brachyolmia
38909645	<i>Clinical management of AI</i>	First-line ceramic crowns; QoL burden; restorative outcomes
23811667	<i>Multidisciplinary management of AI</i>	Multidisciplinary orthodontic/orthognathic/prosthetic care

PMID	Title (abbrev.)	Role in this report
42113459	<i>Condition-specific PROM for AI</i>	Age-dependent QoL impact data
30306291	<i>LTBP-3 and spinal effects on aorta in Marfan</i>	Spinal deformity → aortic mechanics link
40368527	<i>Prenatal diagnosis of Desbuquois dysplasia</i>	Analog reproductive-prevention pathway for AR consanguineous dysplasia

Limitations and Knowledge Gaps

1. **Very small evidence base.** The entire human literature comprises small consanguineous family series (~20+ individuals as of 2020). Precise prevalence, incidence, penetrance quantification, and natural-history data are not available.
2. **Cardiovascular risk quantification.** The magnitude and age-dependence of TAAD risk in biallelic DASS patients (vs. heterozygous carriers) is not established; systematic aortic surveillance data are lacking.
3. **Renal involvement.** Nephrocalcinosis was reported in Moroccan families but its frequency and significance across the disorder are unclear.
4. **Genotype–phenotype granularity.** While the biallelic-LOF vs monoallelic-missense dichotomy is established, finer correlations (which alleles predict aortic risk, dental severity, nephrocalcinosis) are unknown.
5. **No human tissue omics.** Transcriptomic/proteomic/metabolomic profiling of patient tissues has not been reported; mechanistic inference relies heavily on the mouse and on aortic (Marfan-background) studies.
6. **No modifiers or epigenetics.** Disease modifiers and any epigenetic contributions are unstudied.
7. **Context-dependence unresolved.** The paradox that *Ltbp3* removal *ameliorates* aortic disease in Marfan mice while biallelic loss *causes* aortic disease in humans is not fully reconciled.
8. **No therapeutics pipeline.** No disease-specific trials or targeted therapies exist.

Proposed Follow-up Experiments / Actions

1. **Establish an international DASS/*LTBP3* patient registry** to define prevalence, penetrance, expressivity, and natural history, with standardized dental, skeletal, renal, and cardiovascular phenotyping.
 2. **Prospective aortic surveillance study** in molecularly confirmed biallelic and heterozygous *LTBP3* individuals to quantify TAAD risk, define surveillance intervals, and set intervention thresholds.
 3. **Variant functional assays** (latent-complex assembly, fibrillin-1 targeting, SMAD2/3 and ERK1/2 readouts) for all reported variants to refine ACMG classification and genotype–phenotype correlation, including the missense p.Cys1239Tyr.
 4. **Conditional/tissue-specific *Ltbp3* mouse models** (ameloblast-, osteoclast-, and vascular-smooth-muscle-specific knockouts) and hypomorphic knock-ins to dissect branch-specific mechanisms and reconcile the context-dependent aortic paradox.
 5. **Patient-derived models** (iPSC-derived ameloblast/odontoblast organoids; iPSC-derived vascular smooth muscle) to study enamel and aortic pathology in a human genetic background.
 6. **Multi-omics of patient tissues/serum** (transcriptomics, proteomics of ECM/TGF- β components) to identify diagnostic and prognostic biomarkers of aortic risk.
 7. **TGF- β pathway pharmacology screen** in models to evaluate whether pathway modulation (e.g., losartan-type ARBs used in Marfan) mitigates aortic and/or skeletal features.
 8. **Founder-population carrier screening programs** (e.g., Druze Arab communities) coupled with genetic counseling and PGT access to reduce recurrence.
 9. **Prospective QoL/PROM study** specific to DASS integrating dental, growth, and psychosocial outcomes across the lifespan.
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*Report compiled from 17 primary sources across 5 investigative iterations. Evidence source types: human clinical (case/family series), model organism (*Ltbp3*-null and Marfan-cross mouse), in vitro biochemistry (latent-complex/fibrillin studies), and computational (molecular modeling of missense variants).*
